

FAULT CODE 245 - Fan Control Circuit - Voltage Below Normal or Shorted to Low Source

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead and Part Number 3822917 - female Deutsch™/AMP™/Metri-Pack™ test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for an active fault code.	Fault Code 245 active or inactive with more than 1 count in the last 25 engine hours?
STEP 2.	Check the fan control and circuit.	
	STEP 2A. Inspect the fan control and connector pins.	Dirty or damaged pins?
	STEP 2B. Check for an internal short in the fan control.	Within the original equipment manufacturer (OEM) resistance specification?
	STEP 2B-1. Check for a pin-to-ground short circuit in the fan control.	Greater than 100k ohms?

STEPS		SPECIFICATIONS
STEP 3.	Check the ECM and OEM harness.	
	STEP 3A. Inspect the ECM and OEM harness connector pins.	Dirty or damaged pins?
	STEP 3B. Check for a pin-to-ground short circuit in the OEM harness.	Greater than 100k ohms?
	STEP 3C. Check for a pin-to-pin short circuit in the OEM harness or engine harness.	Greater than 100k ohms?
	STEP 3D. Check for an inactive fault code.	Fault Code 245 inactive?
STEP 4.	Clear the fault codes.	
	STEP 4A. Disable the fault code.	Fault Code 245 inactive?
	STEP 4B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check the fault codes.

STEP 1A. Check for an active fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Check for an active fault code. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 245 active or inactive with more than 1 count in the last 25 engine hours? YES	2A
	Fault Code 245 active or inactive with more than 1 count in the last 25 engine hours? NO	Use the following procedure for inactive or intermittent fault code. Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)

STEP 2. Check the fan control and circuit.

STEP 2A. Inspect the fan control and connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the fan control connector from the OEM harness connector.

Action	Specification/Repair	Next Step
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Inspect the OEM harness, engine harness, and ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the fan control or harness connector. Clean the connector and pins. Replace the damaged section of the harness or fan control. Check all harnesses connected in series. Refer to the circuit diagram or wiring diagram for all harness interconnections. • Repair the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html) • Replace the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) • Repair the fan control circuit. Refer to Procedure 019-045 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-045.html)	4A
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	Dirty or damaged pins? NO	2B
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STEP 2B. Check for an internal short in the fan control.

Conditions :

- Turn keyswitch OFF.
- Disconnect the fan control connector from the OEM or engine harness connector.

Action	Specification/Repair	Next Step
Check for low internal resistance of the fan control. • Use a multimeter to measure the resistance between the fan control SIGNAL and fan control RETURN (sensor return 1) pin of the fan control connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Refer to the OEM service manual for resistance specifications. Use the following procedure for connector pin identification. Refer to Procedure 019-360 in Section 19 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Within the OEM resistance specification? YES	2B-1
	Within the OEM resistance specification? NO Repair : Replace the fan control. • Repair the fan control circuit. Refer to Procedure 019-045 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-045.html)	4A

STEP 2B-1. Check for a pin-to-ground short circuit in the fan control.

Conditions :

- Turn keyswitch OFF.
- Disconnect the fan control connector from the OEM or engine harness connector.

Action	Specification/Repair	Next Step
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Short circuit on one connector pin-to-ground check: Measure the resistance and check for a short circuit between the fan control SIGNAL connector pin and ground: Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for connector pin identification. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	2C
	Greater than 100k ohms? NO Repair : Replace the fan control. Repair the fan control circuit. Refer to Procedure 019-045 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-045.html)	4A

STEP 3. Check the ECM, engine harness, and OEM harness.

STEP 3A. Inspect all OEM and engine harness connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness and OEM harness connections.

Action	Specification/Repair	Next Step

Inspect the OEM harness and ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the fan control or harness connector. Replace the damaged section of the harness or fan control. Refer to the circuit diagram or wiring diagram for all harness interconnections. • Repair the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html) • Replace the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) • Repair the fan control circuit. Refer to Procedure 019-045 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-045.html)	4A
	Dirty or damaged pins? NO	3B

STEP 3B. Check for a pin-to-ground in the OEM harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the ECM connector.
- Disconnect the fan control connector from the OEM or engine harness connector.

Action	Specification/Repair	Next Step
Short circuit on one connector pin-to-ground check: • Measure the resistance and check for a short circuit between the fan control SIGNAL pin of the engine harness ECM connector and engine block ground. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for connector pin identification. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	3C
	Greater than 100k ohms? NO Repair : Troubleshoot all harnesses connected in series to determine which contains the pin-to-ground short. Refer to the circuit diagram or wiring diagram for all harness interconnections. • Repair the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html) • Replace the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	4A

STEP 3C. Check for a pin-to-pin short circuit in the OEM harness or engine harness.**Conditions :**

- Turn keyswitch OFF
- Disconnect the engine harness connector from the ECM connector
- Disconnect the fan control connector from the OEM or engine harness connector.

Action	Specification/Repair	Next Step
Short circuit on one connector pin-to-pin check: Measure the resistance and check for a short circuit between the fan control SIGNAL pin of the engine harness ECM connector and all other pins in the connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for connector pin identification. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	3D
	Greater than 100k ohms? NO Repair : Troubleshoot all harnesses connected in series to determine which contains the pin-to-pin short. Refer to the circuit diagram or wiring diagram for all harness interconnections. Replace the damaged section of the harness. - Repair the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html) - Replace the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	4A

STEP 3D. Check for an inactive fault code.

Conditions : - Connect all components. - Turn keyswitch ON. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step

Check for the appropriate circuit response after 30 seconds. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 245 inactive? YES Repair : None. The removal and installation of the connector corrected the fault.	4A
	Fault Code 245 inactive? NO Repair : • Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)	4A

STEP 4. Clear the fault codes.

STEP 4A. Disable the fault code.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Disable the fault code. • Disconnect the fan control solenoid from the OEM or engine harness. • Turn the keyswitch OFF. • Connect the fan control solenoid to the OEM or engine harness. • Turn keyswitch ON. • Use INSITE™ electronic service tool to verify that the fault codes are inactive.	Fault Code 245 inactive? YES	4B
	Fault Code 245 inactive? NO Repair : Reference the Master Repair Manual, Marine C Command™ Basic Panel Manual Systems, Bulletin 4021587, or the Master Repair Manual, Marine C Command Elite™ and C Command Elite Plus™ Panel System, Bulletin 4021617.	Appropriate Troubleshooting Steps

STEP 4B. Clear the inactive fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? YES	Repair complete
	All fault codes cleared? NO Repair : Verify that all steps have been completed. If all steps have been completed, then follow the technical escalation process.	Escalate or call for assistance

